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TECHNICAL DATASHEET

ASCOREPAIR RTM

High Build, High Strength Polymer Modified, Fiber Reinforced, Thixotropic Structural Repair Mortar



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PRODUCT INTRODUCTION

ASCOREPAIR RTM is a high build, high strength, fiber reinforced, thixotropic shrinkage compensated structural repair mortar. ASCOREPAIR RTM is specially engineered for the restoration of spelled and damaged vertical, horizontal and overhead concrete where formwork cannot be utilized. ASCOREPAIR RTM has high, ultimate compressive strength and high abrasion resistance. The specially selected cements, fibers and polymers contained in ASCOREPAIR RTM provide a mortar with strong adhesion to concrete on vertical and horizontal substrates.

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KEY FEATURES

1. High build repairs achievable in a single application
2. Shrinkage compensated allowing for long term dimensional stability
3. Eliminates the need for formwork
4. Shrinkage compensated Exceptional bond strength to concrete substrates
5. Ultimate compressive strength
6. Excellent strength and high abrasion resistance
7. For external & internal use
8. Excellent adhesion with many substrates
9. Internal or external applications
10. Easy to use- simply add water and mix

RECOMMENDED APPLICATIONS

1. Repair of concrete damaged by reinforcement corrosion or damage
2. Repairs to spelled columns, beams, slab etc. caused by corrosion of steel
3. High build repairs for vertical, overhead and horizontal repairs
4. Repairs requiring high compressive strength
5. Repairing damaged concrete panels where structural strength is required
6. Waterproof render to concrete, brickwork and block work.

TECHNICAL PROPERTIES

Appearance	Non-metallic grey powder
Initial Setting Time	4 - 5 hours
Final Setting Time	5 - 7 hours
Fresh Wet Density	2250 ± 50 kg/m^3
Split Tensile Strength (W/P – 0.16)	≥ 3.5 N/mm^2 @ 28 days
Bond Strength (W/P - 0.16)	≥ 2.5 N/ mm^2 @ 28 days
Compressive Strength	≥ 25 N/mm^2 @ 7 days ≥ 40 N/mm^2 @ 28 days

COVERAGE

Approximately 2.5 – 3.5 m² per 30 kg bag at 5-8 mm thickness.

APPLICATION METHODS:

1. Surface Preparation

Substrates should be clean and free from all contaminants, loose particles, coatings, dirt, mould, oil etc. Substrates must be sound, rough and dampened to ensure a good bond. Prior to placing grout, surface should be saturated for a minimum period of four hours and preferably for twenty-four hours. Pre-wet the substrate so that it is slightly moist. Expose the grains close to the surface.

2. Mixing

Pour water into a clean container and add dry mortar ASCOREPAIR RTM. Mix thoroughly with a mixer for approximately 3 minutes until homogeneous. Allow to mature for approximately 1 minute. Then mix again for at least 1 minute until the proper consistency for working has been achieved.

3. Conditions for Use

i. Temperature of the material, air and substrate: from min. + 5 °C to max. + 35 °C
ii. Low temperatures increase, while high temperatures decrease the working and setting time.

4. Application Method

Any cracked or weakened surface should be removed and repaired to provide a solid foundation. It is recommended that for large areas a minimum depth of 5 mm be prepared as to avoid excessive feather edging or skim coating. Break out the repair area to a minimum of 5 mm up to the saw cut edge. Scabbing or high pressure water blasting should be used to remove laitance and provide a mechanical key. If any corroded steel is present remove all loose scale and corrosion/rust deposits. Grit blasting is effective in removing corrosion, and all steel including re-bars should be cleaned to a bright condition. Immediately after cleaning steel, the steel should be treated with ASCOPRIME RUSTCON. This will stop further oxidation and corrosion.

The substrate should be thoroughly soaked with water (any excess being removed) prior to applying one coat of ASCOBOND 450 with OPC at the mixing ratio of 1:1 and applying it well into the surface. ASCOREPAIR RTM can be applied as soon as the ASCOBOND 450 become tacky.

Apply the mixed ASCOREPAIR RTM material to the prepared surface using a trowel or a gloved hand.

Manual working - Use the product to apply a scratch coat. Thickness should be 5 – 25 mm in single layer. If required higher thickness not more than 50 mm, the second layer applied while the first is still wet.

5. TOOLS & CLEAN UP

Compulsory Mixing with twin-shaft mixer. If using a mixing pump, use a batch mixer. Clean tools with soapy water while the material is still fresh.

6. DISPOSAL

Larger quantities of leftover product should be disposed of in the original containers in accordance with the applicable regulations. Completely empty, clean containers should be recycled. Do not dispose of together with household waste. Do not allow to enter the sewage system. Do not empty into drains.

7. Curing

It is a very important to prevent early plastic state moisture loss prior to the product setting. Cure product thoroughly after it has set by keeping it wet and covered for at least 7 days or apply a curing compound.

8. Packaging

Available in 30 Kg and 50 Kg sealed Bag.

9. Shelf Life

6 months from the date of production if stored in original, unopened packaging & in places protected from moisture, sun exposure and frost.

DISCLAIMER:

While the technical details & recommendations contained in this document and the related details given by the representatives of the company correspond to the best of our knowledge & experience, all the above information must in any case be considered as merely indicative and subject to confirmation. Users are recommended to conduct a product suitability test before it is used at full scale. In any case, the consumer alone is entirely liable for any consequences resulting from using the product. For the most up-to-date TDS, please visit our website at www.ascolite.in. Our company policy is one of ongoing R&D; therefore, we reserve the right to update this information without prior notice at any time. As the correct identification of the problems, the quality of other materials used, on-site environmental conditions and the workmanship on-site are factors beyond our control, there is no express or implied guarantee/ warranty as to the results achieved. The company assumes no liability or consequential damage arising from the use of our products for unsatisfactory results. Site visits are not a supervisory responsibility wherever provided. Suggestions made either verbally or in writing by the company may be followed, modified or rejected by the owner, engineer or contractor, since they are solely responsible for carrying out procedures appropriate to a specific application.